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GitHub Link	https://github.com/mrclassic7499/MLA0405-DEEP-LEARNING/tree/main/Assessment/CO5

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 1

Design Desk

COURSE NAME: Deep Learning
SLOT : D

COURSE CODE: MLA0405
Faculty : Dr.Arul Raj A.M

COURSE OUTCOME COVERED

CO5: Students will be able to analyze and explain advanced generative deep learning models including Autoencoders, Deep Belief Networks, Boltzmann Machines, Deep Boltzmann Machines, and Generative Adversarial Networks. They will be able to compare their architectures, explain their learning mechanisms, and apply suitable generative models to real-world problems. (BTL-3)

Course Outcome Weightage: 35%
Assessment Tool Weightage: 30%

Duration: 90 minutes
Mark : 25

Code of Conduct:

I Shaik Mubarak(192425194) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student: Shaik Mubarak

Criteria	Subject Knowledge and Conceptual Understanding (2)	Technical Accuracy and Application of Concepts (3)	Problem-Solving and Analytical Thinking (2)	Communication Skills and Clarity of Explanation (2)	Confidence, Presentation and Professional Behaviour (1)	Total (10)
Weightage	20 %	30%	20%	20 %	10 %	100%
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Faculty Signature :

Total Marks:

1. Design an Autoencoder-Based Feature Learning System

A manufacturing company wants to detect defective products from high-dimensional sensor data. Design an Undercomplete Autoencoder with suitable encoder, latent representation, decoder, activation functions, and reconstruction loss. Explain how the design performs dimensionality reduction and feature learning.

Solution:

AUTOENCODER-BASED FEATURE LEARNING SYSTEM

Problem and Requirement

A manufacturing company collects high-dimensional sensor data such as temperature, pressure, vibration, and speed. The aim is to reduce the dimensionality of this data and identify defective products. The system should learn important features automatically.

Proposed Design

Use an **Undercomplete Autoencoder**.

High-Dimensional Input



Encoder: 128 → 64



Latent Layer: 16



Decoder: 64 → 128



Reconstructed Input

The encoder uses **ReLU activation** and the output layer uses a **Linear activation** for continuous sensor values. The latent layer has fewer neurons than the input, creating a bottleneck.

Working and Training

The model is trained to reconstruct the input using **Mean Squared Error (MSE)**:

$$Loss = \frac{1}{n} \sum (x - \hat{x})^2$$

The encoder compresses the sensor data into a small latent representation containing important features. The decoder reconstructs the original data. A high reconstruction error can indicate an unusual or defective product.

Evaluation and Outcome

Evaluate using **MSE, Precision, Recall, and F1-Score**. The expected outcome is reduced dimensionality, automatic feature learning, and improved defect detection.

2. Design a Robust Regularized Autoencoder

A healthcare organization has noisy patient-record features and needs a robust representation for downstream classification. Design a Regularized Autoencoder using an appropriate regularization strategy. Explain the encoder-decoder structure, regularization mechanism, training objective, and how the design reduces overfitting and improves useful feature extraction.

Solution:

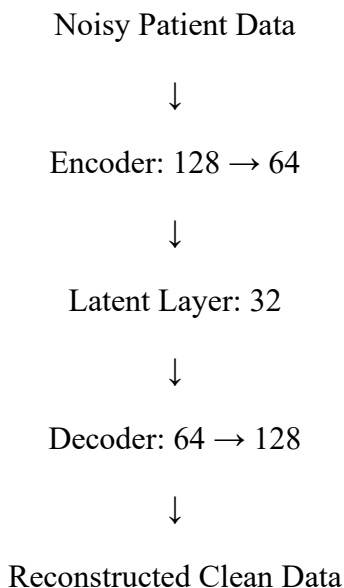
ROBUST REGULARIZED AUTOENCODER

Problem and Requirement

Healthcare patient records may contain noisy and irrelevant features. The aim is to learn robust features for downstream classification while reducing overfitting.

Proposed Design

Use a Denoising Regularized Autoencoder.



The encoder and decoder use ReLU activation. Input noise, Dropout, and L2 regularization are used to improve robustness.

Working and Training

The model receives noisy input but learns to reconstruct the original clean data. The training objective combines reconstruction loss and regularization:

$$Total\ Loss = MSE + \lambda || W ||^2$$

Noise prevents the model from simply memorizing the data, while regularization reduces overfitting.

Evaluation and Outcome

Evaluate using Reconstruction Error, Accuracy, Precision, Recall, and F1-Score. The model produces robust and useful features that can improve downstream classification.

3. Design a Deep Generative Model

A recommendation or content-generation system requires learning complex data distributions. Design a Deep Belief Network or Deep Boltzmann Machine using suitable visible and hidden layers. Illustrate the architecture, explain the learning process, and justify the choice of the model for the proposed application.

Solution:

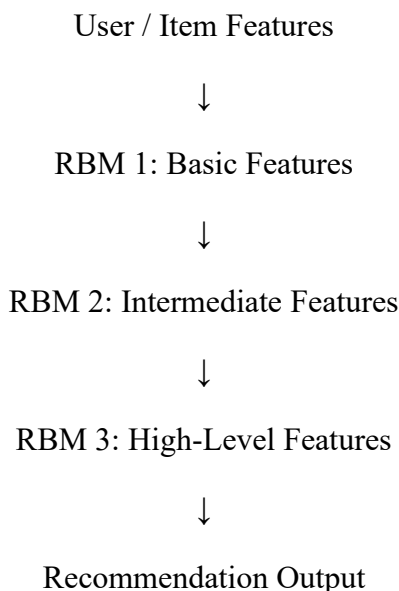
DEEP GENERATIVE MODEL

Problem and Requirement

A recommendation system needs to learn complex relationships between users, items, and preferences. The model should capture hidden patterns and learn high-level representations.

Proposed Design

Use a Deep Belief Network (DBN) made by stacking multiple Restricted Boltzmann Machines (RBMs).



Working and Training

The DBN is trained layer by layer. First, the bottom RBM learns basic patterns. Its output is then used to train the next RBM. After all layers are trained, the complete network is fine-tuned.

This hierarchical learning helps the model capture simple as well as complex relationships in the data.

Evaluation and Outcome

Evaluate using Precision@K, Recall@K, NDCG, and RMSE. The expected outcome is improved recommendations based on meaningful high-level features.

4. Design a Stochastic and Contractive Representation Model

A real-world image or signal-processing application requires representations that remain useful under small input variations. Design a model using Stochastic Encoders/Decoders or a Contractive Encoder. Specify the architecture, objective function, and robustness mechanism, and explain how the design improves representation stability.

Solution:

STOCHASTIC AND CONTRACTIVE REPRESENTATION MODEL

Problem and Requirement

Images and signals may contain small variations due to noise, lighting changes, or measurement errors. The goal is to learn features that remain stable despite these small changes.

Proposed Design

Use a Contractive Autoencoder.

Input Image / Signal



Encoder



Latent Representation



Decoder



Reconstructed Output

The encoder can use CNN/Dense layers with ReLU activation, depending on the application.

Working and Training

The model minimizes both reconstruction error and a contractive penalty:

$$Total\ Loss = Reconstruction\ Loss + \lambda(Contractive\ Penalty)$$

The reconstruction loss preserves important information, while the contractive penalty reduces the sensitivity of the latent features to small input changes.

Evaluation and Outcome

Evaluate using Reconstruction Error, Feature Stability, and Accuracy under noisy inputs. The expected outcome is stable and robust feature representations.

5. Design a GAN-Based Data Generation System

An organization has limited training images and wants to generate realistic synthetic samples. Design a Generative Adversarial Network with a Generator and Discriminator. Specify the input, network flow, adversarial training process, loss functions, and evaluation approach. Compare the proposed GAN design with an Autoencoder and justify why GAN is suitable for the application.

Solution:

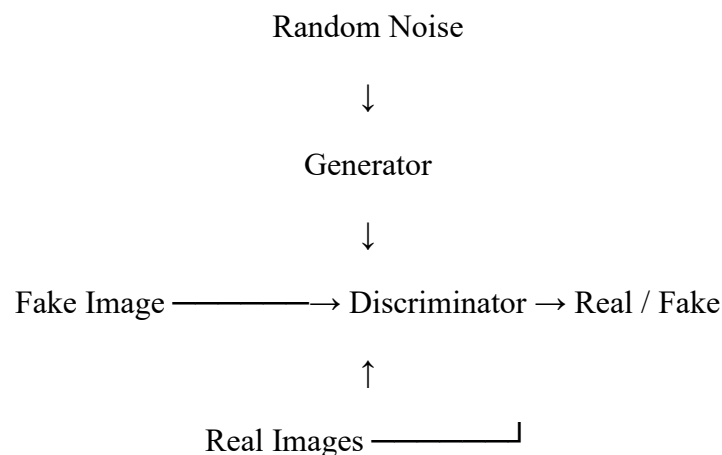
GAN-BASED DATA GENERATION SYSTEM

Problem and Requirement

An organization has a limited number of training images and needs additional realistic samples. The goal is to generate diverse synthetic images to increase the training dataset.

Proposed Design

Use a Generative Adversarial Network (GAN) consisting of a Generator and Discriminator.



The Generator uses Dense and Transposed Convolution layers to create images. The Discriminator uses Convolution layers and a Sigmoid output to classify images as real or fake.

Working and Training

The Generator and Discriminator are trained alternately. The Generator tries to create realistic images, while the Discriminator tries to distinguish real images from fake ones.

$$\min_G \max_D V(D, G)$$

Through this competition, the Generator gradually learns to produce realistic synthetic images.

Evaluation and Outcome

Evaluate using FID Score, Inception Score, image diversity, and visual quality.

Unlike an Autoencoder, which mainly reconstructs and compresses input data, a GAN can generate completely new realistic samples. Therefore, GAN is suitable for increasing limited image datasets.

Design Desk Guidelines

- Identify the real-world problem, requirements, inputs, outputs and constraints.
- Draw a clear architecture or workflow diagram for the proposed model.
- Specify important layers, units, activation functions, latent representation and loss functions.
- Explain the training or learning mechanism and justify major design choices.
- Mention suitable evaluation measures and expected outcomes.
- Use neat labeling and present the design in a logical sequence.
- The solution should be original and written in the student's own words.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Understanding of Problem Context	20	Clearly understands the problem, requirements, constraints, and expected outcomes.	Good understanding with minor gaps in requirements.	Basic understanding; some requirements are unclear.	Poor understanding or misinterpretation of the problem.
Application of Concepts	30	Accurately applies appropriate generative deep learning concepts and architectures.	Applies relevant concepts correctly with minor errors.	Applies basic concepts with limited accuracy.	Fails to apply appropriate concepts.
Design Approach	20	Innovative, logical, well-structured, feasible, and technically justified design.	Logical and feasible design with minor gaps.	Basic design with some inconsistencies.	Unclear, illogical, or impractical design.
Accuracy of Solution	20	Completely correct architecture, process,	Mostly correct with minor technical errors.	Partially correct solution with noticeable gaps.	Incorrect or incomplete solution.

		objectives, and justification.			
Clarity	10	Clear, well-organized diagram and explanation with proper technical labeling.	Understandable with minor clarity issues.	Limited clarity and explanation.	Poorly presented and difficult to understand.